IMAGE & VIDEO PROCESSING · 2026-27 · PRESENTATION + IEEE PAPER

NavCert Offline, the IVP cheat sheet

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IF YOU REMEMBER ONE SENTENCE

A phone photo of a certificate has four defects. Classical image processing fixes the pixels before any neural network reads them, on the phone, with no internet, and straightening the page is the step that matters most: OCR error falls from 9.6% to 4.2%.

THE STORY IN FIVE BEATS

The flow you are walking them through

Slides 1-3

The problem

A crooked, shadowed, blurry photo with a stamp on it. OCR wants a flat, evenly lit page.

Slides 4-6

The idea

Fix the pixels first. Two-stage pipeline, mapped to all 8 syllabus units.

Slides 7-11

The techniques

Geometry, light, noise, thresholds and colour, then motion for the video unit.

Slides 12-13

The proof

Measured on 200 paired captures, then shown running in the real app.

Slides 14-15

The honesty

What is hand-tuned and synthetic, what comes next, callback close.

SLIDE BY SLIDE

What to say, and what you must understand

01

20 s

TITLE

Crooked photo + classical IVP = a certificate you can search

> *"NavCert Offline turns a phone photo of a seafarer's certificate into clean, searchable text and five fields, with no network at all. Almost every unit of this course is doing real work inside it."*

UNDERSTAND IT

The five fields are certificate name, issue date, port of issue, issuing authority and expiry date. "Offline" means the whole chain runs on the phone: the app has no internet permission.

02

35 s

THE PHOTO

4 defects. 1 photo. 0 bars of signal.

> *"Perspective skew, uneven light and shadow, blur and sensor noise, and a red stamp over the text. And on a ship there is often no signal."*

UNDERSTAND IT

These four defects are the spine of the whole talk. Each later technique slide fixes one of them: geometry fixes skew (slide 7), histogram processing fixes light (8), filters fix noise (9), segmentation and colour separate ink from stamp (10).

If asked: why a synthetic certificate?
The real certificates are a client's private documents. Every name on the sample is invented. The IMO source is MSC.1/Circ.1665 (2023), which assumes the verifier has internet.

03

35 s

THE PROBLEM

The OCR is ready. The photo isn't.

> *"How much of that gap can classical image processing close before any neural network sees a pixel?"*

UNDERSTAND IT

OCR models are trained on upright, evenly lit, sharp text. A phone gives projective distortion, an illumination gradient, noise, blur and JPEG blocking. ML Kit's own guide asks for characters of at least 16×16 pixels.

If asked: why not just a bigger model?
Labelled certificate data is small, and a deterministic fix can be inspected stage by stage.

04

30 s

THE IDEA

Fix the ~~model~~ pixels first

> *"Classical image processing is deterministic, explainable at every stage, and needs no training data. The neural models then do what they're good at: reading clean text."*

UNDERSTAND IT

This is the thesis of the deck and the paper. The six-stage strip on the slide goes raw photo → edges → detected quadrilateral → warped page → CLAHE → adaptive black-and-white.

05

45 s

THE PIPELINE

A dry dock for documents

> *"Row one finds the page on a 1600-pixel copy. Row two cleans the page at up to 3200 pixels. Only then does OCR run. Every number on this slide is a constant in DocumentScanner.kt."*

UNDERSTAND IT

Row 1, find: grayscale → Gaussian 5×5 → Canny 55/170 → close 5×5 → largest convex quadrilateral ≥15% of frame. The quality gate runs alongside: mean brightness <55 is too dark, >220 too bright, Laplacian variance <65 is blurry.

Row 2, clean: perspective warp → CLAHE (clip 2.2, 8×8) → median 3×3 → unsharp mask (Auto/Clean) or adaptive threshold 35/11 (B&W) → 18 px white border → OCR.

If asked: why two resolutions?

Finding the border needs structure, so a small copy is faster. OCR needs stroke detail, so it gets the larger image. The app only downsamples, it never upscales.

06

40 s

SYLLABUS MAP

8 units. 1 scan.

> *"The same pipeline as a route through the syllabus. Solid stops are the app's capture path; dashed stops run in the Pipeline Lab on the user's own scan."*

UNIT	WHERE IT LIVES	PATH
1 Fundamentals	Resolution, quantisation, warp, distance	App + Lab
2 Enhancement	CLAHE, median, unsharp, point ops, DFT low-pass	App + Lab
3 Transforms	JPEG (8×8 DCT) for saved previews, DCT demo	App + Lab
4 Morphology	Closing the Canny border; open, close, boundary	App + Lab
5 Edges	Canny; Sobel, Laplacian, LoG; Laplacian blur check	App + Lab
6 Segmentation	Adaptive threshold; Otsu; connected components	App + Lab
7 Colour	RGB/HSV/YUV channels, hue slicing of the stamp	Lab
8 Video	Block matching ES / TSS / 2-D log, I/P/B frames	Lab

07

50 s

EDGES · MORPHOLOGY · GEOMETRY

Canny + closing + 4 corners = a flat page

Units 1, 4, 5, 6

> *"Four corners give eight equations, exactly the eight unknowns of a homography."*

UNDERSTAND IT

Canny: smooth, take the gradient, thin edges to one pixel (non-maximum suppression), then hysteresis. An edge must start above 170 but may continue down to 55, so the page border survives through the shadow.

Closing (dilate then erode) bridges small gaps in the border so the contour is closed.

Contours → polygon (approxPolyDP, ε = 2% of perimeter). Keep the biggest convex four-sided shape.

Homography: a 3×3 matrix with 8 unknowns maps any quadrilateral to a rectangle. Each corner gives 2 equations (x and y), so 4 corners give 8.

NUMBERS

`page found 184/200`

`IoU ≥ 0.9 on 156/200`

`median corner error 1.3 px`

If asked: what if no page is found?
The app keeps the whole image and offers manual crop. All 16 misses were low-light captures where Canny finds no usable border.

08

45 s

HISTOGRAM PROCESSING

64 dimmer switches, not one

Unit 2, Unit 1

> *"Global equalisation is one dimmer switch for the whole page. CLAHE is 64 of them, an 8×8 grid of tiles, each clipped so it can't amplify noise."*

UNDERSTAND IT

Histogram equalisation maps each gray level through the cumulative histogram (CDF) so levels spread across 0–255. One global mapping is dominated by white paper, so a shadowed corner stays dark.

CLAHE does this per tile. It clips each tile's histogram at 2.2× the average bin and redistributes the excess, which caps contrast gain and therefore noise. Neighbouring tiles are blended bilinearly so there are no seams.

Key nuance to remember
Enhancement is the smaller step. Word recall barely moves (92.9% raw vs 93.6% enhanced). Most of the gain comes from rectification restoring line order.

09

45 s

SPATIAL · FREQUENCY · DCT

Blur it to sharpen it

Unit 2, Unit 3

> *"Subtract a blurred copy and what's left is edge detail. Add some of it back. And a median beats averaging on speckle because it ignores outliers."*

UNDERSTAND IT

Unsharp mask: out = I + (a − 1)(I − G_σI). I − blurred I is the high-frequency detail. Auto adds 32% of it back, Clean 50%.

Median vs mean: salt-and-pepper pixels are outliers. The median of a 3×3 window simply drops them; a mean smears them into neighbours.

DFT: blurring is a low-pass filter. Text strokes live in the high frequencies, which is why over-smoothing hurts OCR.

DCT: JPEG splits the image into 8×8 blocks and keeps the few coefficients that carry most of the energy (energy compaction). Every saved preview is JPEG quality 88.

IN THE APP, SAMPLE CERTIFICATE (PSNR)

`noisy 17.11 dB` `median 31.18 dB`

`Gaussian 24.58 dB` `mean 24.26 dB`

`JPEG q10 30.13 · q50 37.74 · q90 45.28 dB`

BENCHMARK

`q10 keeps 4.7% of coefficients`

`PSNR 27.7 dB`

`OCR CER still 0.35%`

10

50 s

SEGMENTATION · COLOUR

No single threshold survives a shadow

Units 4, 6, 7

> *"Otsu is optimal when the histogram has two clean peaks. A shadow smears the paper into a wide band, so no single threshold works. Adaptive compares each pixel with its own neighbourhood."*

UNDERSTAND IT

Otsu tries every threshold T and picks the one that maximises between-class variance (ink vs paper as separated as possible). In the app's Lab, Otsu picks T = 175 and the shadowed corner turns black.

Adaptive: pixel is ink if it is darker than the Gaussian mean of its 35×35 neighbourhood minus 11. The local mean follows the lighting.

Connectivity: 4-connectivity breaks diagonal strokes; 8 keeps characters whole (463 vs 442 components on the sample).

Colour slicing: in HSV, hue separates the red stamp from black ink regardless of brightness. Red wraps around hue 0, so use two bands.

SHADOW CAPTURES, N = 24 (OCR CER)

`global 34.2%` `Otsu 39.2%`

`adaptive 0.05%`

If asked: is 0.05% realistic?
It is the isolated case: the page was rectified with true corners, so only thresholding is being tested.

11

45 s

VIDEO · MOTION ESTIMATION

Dead reckoning for pixels

Unit 8

> *"The camera preview is a video. For each block in the new frame, search the old frame for the best match. Exhaustive tests 225 positions; three-step tests 25."*

UNDERSTAND IT

Block matching: 16×16 blocks, search window ±7 px, cost = sum of absolute differences (SAD). The best offset is the motion vector.

Exhaustive: $(2p + 1)^2 = 15^2 = 225$ positions. **Three-step:** 9 + 8 + 8 = 25, step size 4 → 2 → 1. **2-D log:** a "+" pattern; step halves when the centre wins.

I/P/B frames: I is coded alone; P is predicted from a past frame with motion vectors + a DCT residual; B uses past and future and is usually smallest.

In the app this is a Lab demo: near-zero vectors mean a steady hand. Capture itself is a single still photo.

NUMBERS

`points/block 225 · 25 · 20.7`

`in app 212.9 · 24.4 · 20.0`

`shake detected (−4, 3) px`

The honest trade-off

On textured text blocks, exhaustive finds the exact vector 99.9% of the time, three-step 29.6%, 2-D log 32.8%. Text repeats, so the error surface has many local minima and fast searches get trapped.

12

50 s

MEASURED EVIDENCE

One page, three photos, one ruler

> *"Same synthetic certificates, known ground truth, same OCR engine. Raw 9.6%, rectified 4.2%, enhanced 4.2%. Rectification does the heavy lifting; enhancement adds nothing on average, and I'd rather show that than hide it."*

UNDERSTAND IT

CER (character error rate) = (substitutions + deletions + insertions) ÷ characters in the truth. It is edit distance per character.

Why rectification wins: on a tilted page the OCR still reads most words, but lines interleave, so a label like "Date of expiry" no longer sits on the same line as its value. Straightening restores reading order.

Tilt alone: raw CER 20.3% → 0.4% after rectification. Enhancement helps in low light and blur, hurts slightly in overexposure.

HEADLINE

`CER 9.6% → 4.2%`

`5 fields exact 69.1% → 88.4%`

`paired drop 5.4 pts, 95% CI 3.9–7.1`

`better 112 · worse 40 · tied 48`

`B&W mode 7.6% CER`

Setup in one line

40 synthetic pages, 200 captures, 8 degradations + clean, PP-OCRv6 small on a desktop CPU, fixed seeds. It measures what image processing contributes, not accuracy on real certificates.

13

50 s

ON THE PHONE

"Image looks good."

> *"This is the real app. The status line says 'Image looks good, page borders found and aligned'. Auto, Clean and B&W are the three modes from slides 8 to 10. Then OCR and the five fields, all on the device."*

LIVE DEMO PATH (IF MIRRORING WORKS)

Airplane mode on and say so → Choose an image → sample certificate → show the status line → toggle Auto / Clean / B&W → Extract text → five fields → Open Pipeline Lab for this page → swipe histogram and threshold stages. Never wait more than 5 seconds; fall back to the screenshots.

Note

The port field now reads "Port of Valmora" correctly. If the slide notes mention the port repeating the authority, that bug is fixed.

14

35 s

LIMITATIONS · FUTURE WORK

Works on a phone. Not yet proven on a fleet.

> *"Thresholds are hand-tuned, quality is measured on the whole frame, contours need a visible border, and my error rates are on synthetic pages. Next: calibrate on labelled phone photos and measure on real certificates."*

If asked: which syllabus topics are not built?

Hadamard transform, hit-or-miss, thinning and thickening, gray-scale morphology, region growing and split-and-merge, Bayesian/EM binarisation. A page is one large region with a strong border, so contours are cheaper and more reliable than region growing.

15

25 s

THANK YOU · QUESTIONS

4 defects in. 1 clean page out. Still 0 bars.

> *"Edges and morphology fixed the geometry, histogram processing fixed the light, filters fixed the noise, segmentation and colour separated ink from stamp, and it still runs with zero bars. Thank you."*

WHY THIS CLOSE WORKS

It is a callback to slide 2: the same four defects, each now matched to the unit that fixed it. Stop talking and step back.

FORMULAS AND ONE-LINE MEANINGS

The maths behind each stage

CONCEPT	FORMULA / RULE	WHAT IT MEANS
Grayscale	$Y = 0.299R + 0.587G + 0.114B$	Luminance carries legibility; one channel is a third of the data.
Gaussian kernel	$G(x,y) \propto \exp(-(x^2+y^2)/2\sigma^2)$	Weighted blur that removes paper grain before edges.
Canny hysteresis	start > 170, continue > 55	Strong edges seed, weak connected edges survive. Ratio about 1:3.
Closing	$A \bullet B = (A \oplus B) \ominus B$	Dilate then erode: fills gaps without growing the shape.
Homography	$[x'\ y'\ w']^T = H\ [x\ y\ 1]^T$, 8 DOF	4 corners × 2 coordinates = 8 equations.
Focus measure	$\mathrm{var}(\nabla^2 I) < 65 \rightarrow$ blurry	Sharp images have strong second derivatives, so high variance.
Histogram eq.	$s = (L-1) \cdot \mathrm{CDF}(r)$	Spreads gray levels; CLAHE does it per tile with a clip limit.
Unsharp mask	$\mathrm{out} = I + (a-1)(I - G_\sigma I)$	Adds back edge detail. Auto a = 1.32, Clean a = 1.50.
Otsu	$T^* = \arg\max\ \omega_0\omega_1(\mu_0-\mu_1)^2$	Threshold that best separates two classes.
Adaptive threshold	ink if $I <$ mean$_{35\times35} - 11$	Each pixel judged against its own lighting.
2-D DCT	8×8 block → 64 coefficients	Energy compaction; JPEG quantises high frequencies.
PSNR	$10 \cdot \log_{10}(255^2 / \mathrm{MSE})$	Higher dB = closer to the original.
Block matching	$\mathrm{SAD} = \Sigma\lvert B_t - B_{t-1}(\mathrm{shifted})\rvert$	ES $(2p+1)^2 = 225$; TSS 25; 2-D log ≈ 20.
CER	(S + D + I) / N	Edit distance per reference character.

MEMORISE THESE

The numbers you will be asked about

NUMBER	WHAT IT IS	SOURCE
9.6% → 4.2%	Mean OCR character error rate, raw photo → rectified	Benchmark, n = 200
69.1% → 88.4%	All five fields exactly right	Benchmark, 1,000 field slots
5.4 pts	Paired CER drop, 95% CI 3.9–7.1	Bootstrap
184 / 200	Page found; 156 with IoU ≥ 0.9; median corner error 1.3 px	Benchmark
39.2% vs 0.05%	Otsu vs adaptive threshold CER under cast shadow	Benchmark, n = 24
0.35%	OCR CER after DCT at quality 10 (4.7% coefficients kept)	Benchmark
225 · 25 · 20.7	Search points per block: exhaustive, three-step, 2-D log	Benchmark
99.9 · 29.6 · 32.8%	Exact motion vector on textured blocks, same order	Benchmark

T = 175	Otsu threshold on the sample certificate	In app
31.18 dB	Median filter PSNR on salt-and-pepper (mean 24.26)	In app
~37 ms	Classical Auto path per capture vs ~1063 ms OCR (desktop)	Benchmark timing

THE IEEE PAPER

How the paper lines up with the deck

Title: **NavCert Offline: Classical Image Processing for Robust On-Device Digitization of Maritime Certificates**. 8 pages, IEEE two-column, 8 figures, 8 tables, 14 equations, 41 references.

I · INTRODUCTION

Slides 1–3

Maritime problem with sources: about 2.57M STCW seafarers on 85,148 ships (BIMCO/ICS 2026); Tokyo MoU crew-certificate deficiencies 877 → 1,304 (2022 → 2024); IMO e-certificate guidance assumes internet. Three contributions: offline pipeline, paired benchmark, Pipeline Lab.

II · RELATED WORK

Background for Q&A

OCRdroid (needed a server), ICDAR 2015 SmartDoc (phone capture), Canny, Otsu, Sauvola, CLAHE, PP-OCR, ML Kit, LayoutLM, Donut, MobileBERT, MiniLM.

III-IV · SYSTEM AND PIPELINE

Slides 4–5, 7–10

Architecture and every stage with its equation and constant.

V · PIPELINE LAB

Slides 6, 8–11

The in-app tool that runs 11 syllabus stages on the user's own scan.

VI · RECOGNITION AND FIELDS

Slide 13

ML Kit for print, PP-OCRv5 for handwriting, rules → fine-tuned MiniLM → MobileBERT for the five fields. MiniLM got 15/26 on the human-checked gold set.

VII-IX · SETUP, RESULTS, DISCUSSION

Slide 12

Synthetic benchmark design, all results tables, and why rectification dominates, when enhancement helps or hurts, why Otsu fails, why fast block matching fails on text.

X-XI · LIMITS AND CONCLUSION

Slides 14–15

Synthetic data, desktop OCR engine instead of ML Kit, no phone latency yet, hand-tuned thresholds. Next: real labelled certificates on the phone.

ONE-LINE ABSTRACT

Say this if asked "what is your paper?"

"An offline Android pipeline whose classical front end halves OCR error on phone-captured certificates, measured on 200 paired synthetic captures, mostly by geometric rectification."

Questions you should be able to answer in two sentences

Why Canny thresholds 55 and 170?
Hand-tuned on capture tests, roughly the recommended 1:3 low-to-high ratio. Hysteresis lets the border continue through the shadow.

Why CLAHE instead of global histogram equalisation?
Global HE uses one mapping dominated by white paper, so shadows stay dark. CLAHE uses 64 clipped local mappings, so a shadow in one corner doesn't affect the rest and noise gain is capped.

Why does Otsu fail under a shadow?
It assumes a two-peak histogram. A shadow shifts the background level so paper spans a wide band, and no single T separates ink from paper everywhere.

How do four corners give a homography?
A homography has 8 degrees of freedom. Each corner correspondence gives 2 equations, so 4 corners solve it exactly.

Median or Gaussian for salt-and-pepper noise?
Median. Impulse pixels are outliers; the median discards them while a weighted average spreads them. 31.18 dB vs 24.58 dB in the app.

What does Laplacian variance measure?
Focus. The Laplacian responds to rapid intensity change; a blurred image has weak edges, so its Laplacian has low variance. Below 65 the app warns.

Why grayscale before OCR?
Legibility is carried by luminance contrast, and one channel is a third of the data. Colour is only needed for stamp slicing.

Why an 18-pixel white border?
Text detectors can miss boxes that touch the image edge. The margin keeps edge characters inside.

Why is three-step search less accurate?
It assumes the error surface falls steadily toward the true match. Text is repetitive, so there are many local minima and the coarse first step can land in the wrong one.

What are I, P and B frames?
I is coded on its own. P is predicted from a previous frame using motion vectors plus a DCT-coded residual. B uses past and future frames and is usually the smallest.

Why synthetic data?
The real certificates are private client documents, and synthetic pages give exact ground-truth text, fields and corners, so the effect of each stage can be measured cleanly.

Did enhancement help?
Not on average: rectified and enhanced both score 4.2%. It helps in low light (11.1% → 7.2%) and blur, and hurts in overexposure. I report that as a finding.

SAY THESE PLAINLY

Honest limits, and what you never claim

Limits to admit

- Canny, brightness and blur thresholds are hand-tuned.
- Quality is measured on the whole frame, before rectification.
- Contours need a visible page border; corner ordering is weak near 45°.
- CER is measured on synthetic pages with PP-OCR on a desktop, not ML Kit on the phone.
- Phone latency and memory have not been measured.

Never claim

- **Not** "88% accuracy on real certificates". It is a synthetic benchmark.
- **Not** that motion estimation runs in the capture path. It is in the Pipeline Lab.
- **Not** that enhancement improved OCR on average. Rectification did.
- **Not** that colour mode exists in capture. All three modes output grayscale.